GENAI.WORKS OPEN AGENT HACKATHON 2026

PeoplePay-360 · PayProof

Proof-carrying commission dispute resolution. It works out what should have been paid, shows the evidence, tests the alternatives, and only then acts, with approval.

Synthetic data · simulated payroll adapter · Python stdlib + vanilla JS

PROBLEM

“Why is my commission wrong?” has five different answers

CRM
$60,000 opportunity

Contract
amended to $50,000

Billing
partly paid, credit note, payment recorded late

Comp plan
collection basis, versioned rates, exceptions

Payroll
$3,600 paid, then the same clawback twice

Analysts reconcile this by hand. Chatbots guess. Neither leaves a proof, and neither is safe to let near payroll.

Target users: sales-compensation and payroll operations analysts, finance approvers, and the reps who file disputes.

DEMO CASE · CASE-1001

Claimed $3,000. Verified $1,100.

Rep expectation: CRM $60,000 × 100% × 10%	$6,000.00
Amendment AMD-3301-1 → eligible $50,000 (P-3.1)	−$1,000.00
Only $43,000 net collected by cutoff (P-3.2)	−$700.00
VERIFIED DUE	$4,300.00
Paid: $3,600 − $200 − $200 (duplicate clawback)	$3,200.00
VERIFIED UNDERPAYMENT	$1,100.00

Every line cites source records and a policy clause. All of it is computed by deterministic Decimal code.

HOW IT REASONS

An adversarial workflow with a deterministic verifier

Investigator → Skeptic → Policy verifier → Resolution planner → Final validator

Skeptic's strongest counterargument
“PAY-44987 was recorded after cutoff. Exclude it and she's owed $200.” Tested with replay, then ruled out under P-4.2 (value date governs).

Materiality, not vibes
Conflicting amendments are recomputed under each alternative. If the outcome changes the case is DISPUTED; if not, it is VERIFIED.

Model at the edge
Nemotron only phrases explanations. Any sentence with an unknown record id or an invented amount is dropped and counted.

SIGNATURE FEATURE

Counterfactual payout replay

HYPOTHETICAL SCENARIO	DUE	Δ VARIANCE
Timing by recorded date instead of value date	$3,400.00	−$900.00
Invoice paid in full before cutoff	$4,800.00	+$500.00
Credit note issued after cutoff	$4,500.00	+$200.00
Duplicate clawback never posted	$4,300.00	−$200.00
Amendment void (CRM value governs)	$4,300.00	$0.00

Same engine, hypothetical overlays, always labelled. The rep's $60k premise turns out not to matter, because collections cap the base.

SAFE ACTION

Approval gated, executed at most once, verified afterwards

PROPOSED → VALIDATED → AWAITING_APPROVAL → APPROVED → EXECUTED → VERIFIED

- Backend authorization, approval limits ($5k approver / controller above), segregation of duties. The rep's approval gets a 403.
- Idempotency key and source fingerprint are re-checked at approval and at execution. Stale evidence moves the action to EXPIRED.
- Postcondition: the case is re-investigated with the simulated ledger, so paid = $4,300 and variance = $0. A hash-chained receipt is issued.
- Negative case: no contract means NEEDS_EVIDENCE, and the only action allowed is to request evidence.

EVALUATION · ACTUAL RESULTS

25 hand-labelled synthetic cases

METRIC	PAYPROOF	RULES-ONLY BASELINE
Decision accuracy (incl. amount when money moves)	25/25	11/25
Wrong payroll adjustments recommended	0	10
Correct abstention when evidence is missing or contradictory	5/5	0/5
Citation coverage of material claims	100%	0%
Unauthorized actions succeeded / attempts	0 / 151	n/a
Forced postcondition failures that still produced a receipt	0 / 25	n/a
Forced postcondition failures that still produced a receipt	0 / 25	n/a
Duplicate actions from 125 concurrent re-executions	0	n/a
Stale-evidence executions blocked	8 / 8	n/a

Limitations: the cases and the engine were written by the same team, so these results show policy fidelity, not generalisation. The baseline is a transparent strawman. The model ran in deterministic-fallback mode, so there are no live-model results.

HOW ZETARIS AND METERLESS ARE USED

Meterless World Model as the evidence backbone

1 · Evidence intake
Every source record is ingested with provenance (source system, recorded time) and bitemporal edges (validFrom = effective time).

2 · Verifier cross-check
The World Model's as-of-cutoff view must contain exactly the cash events the Decimal engine counted. Otherwise the proof is REJECTED.

3 · Context builder
The Nemotron prompt is built from the World Model neighbourhood of the disputed deal.

4 · Stale-evidence gate
Each action snapshots the World Model. If re-ingest reports `changed=true` at approval or execution, the action EXPIRES.

Zetaris: one fixed, validated SELECT per source system over the documented REST SQL route, with per-system provenance on every investigation.

What's unique: the bitemporal edges carry the core payroll distinction, value date versus recorded date. The engine's idempotency contract becomes a payroll safety control.

SPONSOR TECHNOLOGY — TRUTHFUL STATUS

What is live, and what isn't yet

Meterless World Model	LIVE in-process on every investigation. A one-to-one port of the official 8-check conformance suite passes 8/8.
Zetaris	Implemented to the documented REST route; not yet run against a live instance. Stub-contract-tested; live smoke test included.
NVIDIA Nemotron	Client for integrate.api.nvidia.com implemented; not yet called with a real key. Output is cite- and amount-checked.
Cursor	Open item: this build was produced in Claude Code. The team's Cursor use must be described truthfully.

The UI shows LIVE / CONFIGURED_UNVERIFIED / FALLBACK / UNAVAILABLE from runtime outcomes. A mock never reads LIVE.

TECHNICAL STACK

Small, reproducible, judge-runnable

Core
Python standard library only, `decimal` money, an event-sourced Meterless World Model, and a stdlib HTTP API

UI
Vanilla JS Decision Room, no build step, responsive

Data
Zetaris REST SQL or local synthetic fixtures (SAMPLE_MODE)

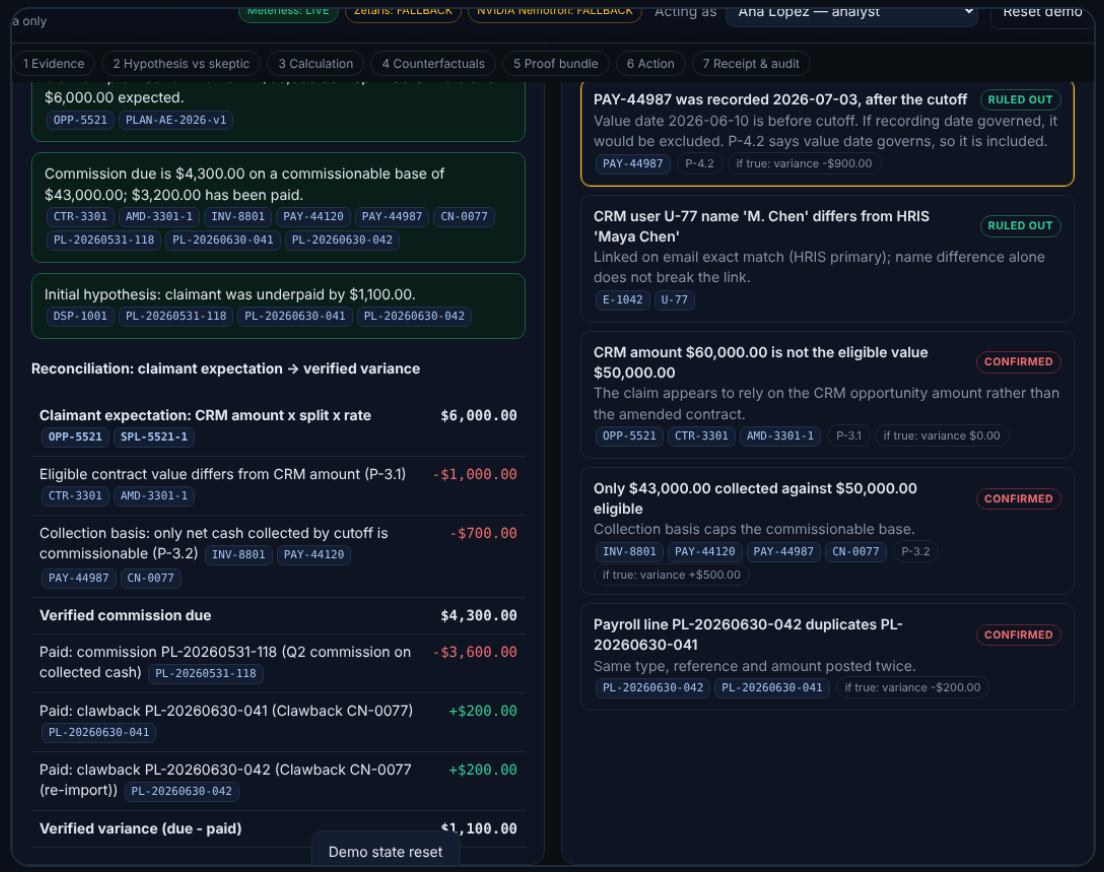
Model
NVIDIA Nemotron via OpenAI-compatible API, at most 2 calls per run, cached

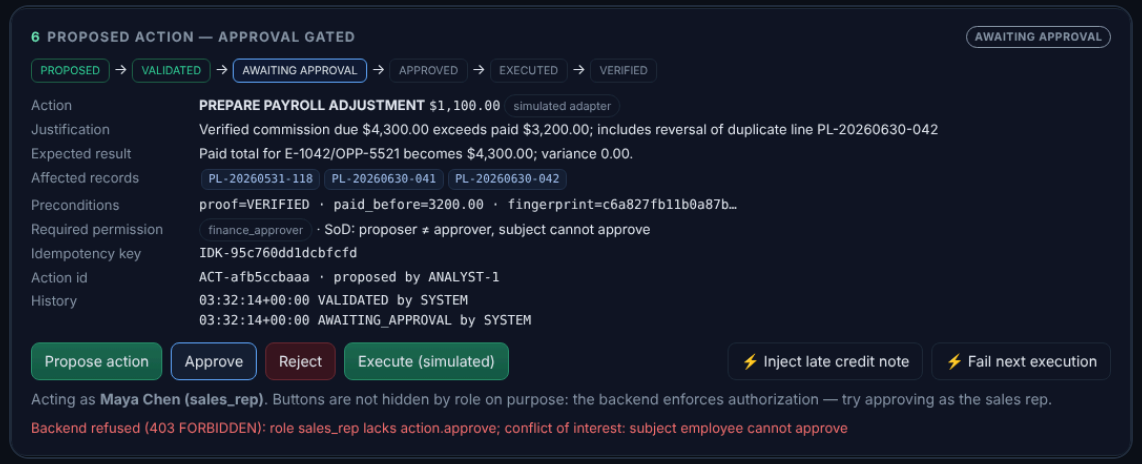
Evidence
JSONL traces in `logs/`, 3 samples, 44 tests, a 25-case benchmark

Run
`python3 run.py` or Docker; `run.py --check-integrations`

PRODUCT VISUALS

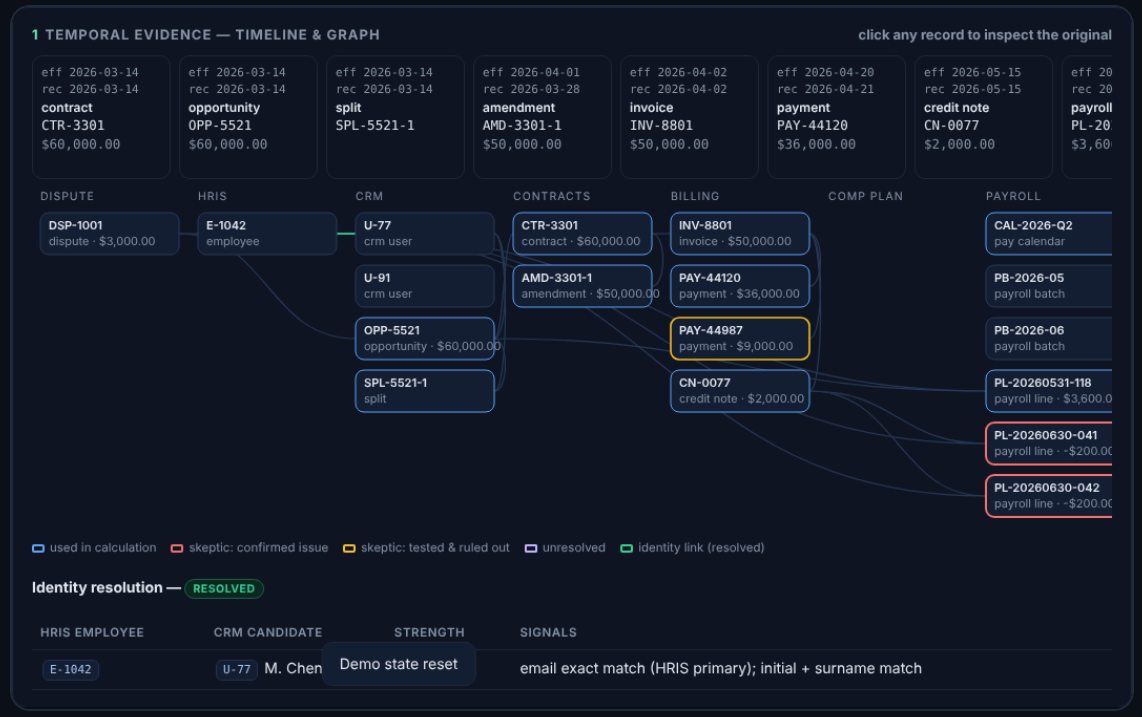
The Decision Room

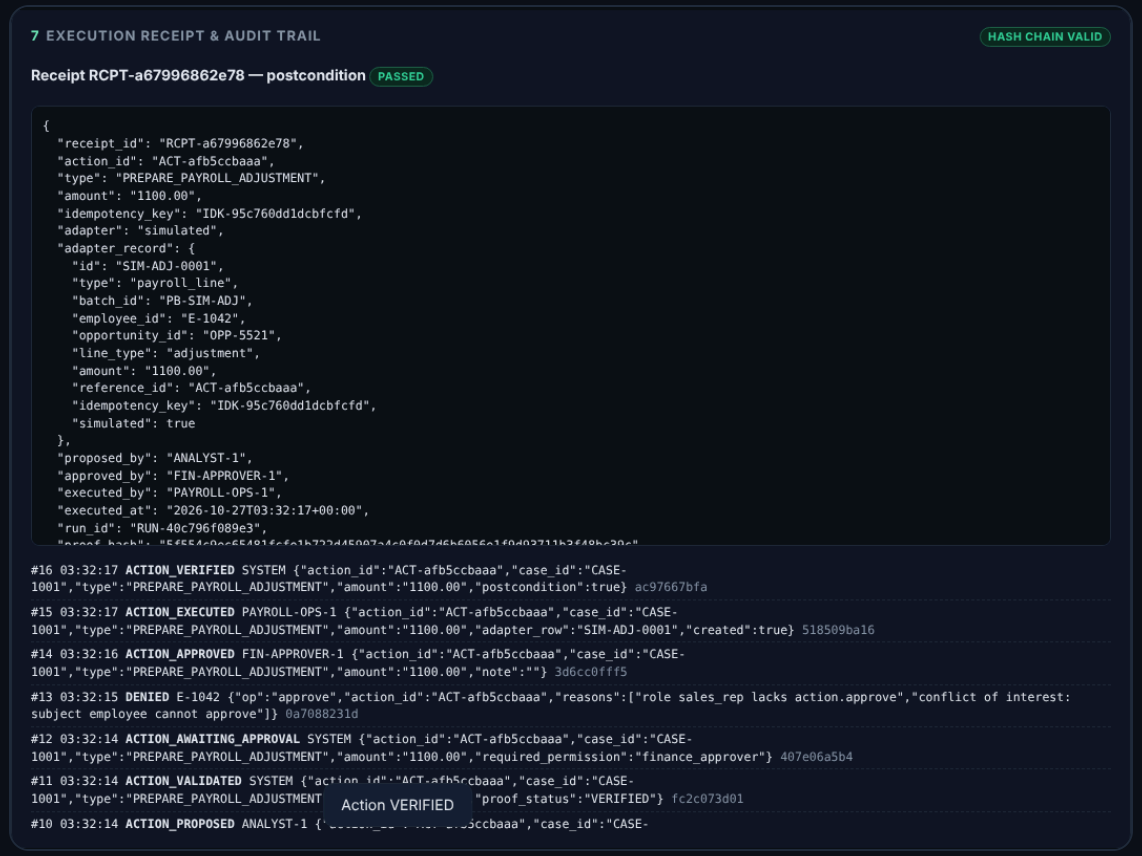




PRODUCT VISUALS

Evidence graph and execution receipt





FUTURE ENHANCEMENTS

Next

- A live Zetaris semantic layer over real HRIS / CRM / billing / payroll connectors
- Meterless H-MEM memory of prior rulings per plan clause, so precedent is cited in new disputes
- A durable event store, SSO with role mapping, and real payroll adapters behind the same gate
- Multi-plan support: accelerators, quotas, multi-period clawbacks
- An independent, externally labelled benchmark set

TRY IT

python3 run.py

No dependencies and no credentials. It opens on CASE-1001 at http://127.0.0.1:8360.

github.com/victrvondoom/Open-Agent

python3 -m unittest discover -s tests · python3 -m payproof.bench